SokoRadar — Capstone Project Proposal

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Project at a glance

SokoRadar forecasts staple food prices in Kenyan markets one to three months ahead and warns farmers, traders, county officials and aid agencies before a price shock hits — through a web dashboard, an open API and a USSD menu that works on any phone.

Item	Summary
Working title	SokoRadar: an AI early-warning system for staple food price shocks in Kenya
Problem	Price spikes in maize, beans and maize flour hit poor households and pastoralists hardest, yet forward-looking, market-level warnings rarely reach the people who act on them
Solution	Forecast prices per market and commodity, turn forecasts into Normal / Stress / Alert / Crisis warnings, and explain what drives each warning
Data	WFP market prices, CHIRPS rainfall and MODIS vegetation indicators (all on HDX), KES/USD rates (CBK), global commodity prices (World Bank)
Methods	Time series forecasting (SARIMAX, Prophet, LightGBM), alert classification, conformal prediction intervals, SHAP explanations, backtests on past crises
Deployment	Streamlit dashboard, FastAPI service, Africa's Talking USSD and SMS channel, monthly auto-refresh with GitHub Actions
Reach	Starts with WFP-monitored Kenyan markets, grows toward all 47 counties through KAMIS, and runs on WFP data for neighbouring countries

Background and problem statement

Kenya's hunger crisis reaches households through prices: 2.7 million people in the arid and semi-arid lands (ASALs) faced Crisis-level food insecurity (IPC Phase 3+) in July–October 2026, up 55% on the previous analysis, with 348,000 in Emergency (Capital FM, Sep 2026).

Prices are rising faster than incomes. Food inflation hit 9.5% in September 2026, while headline inflation rose to 6.8%, its highest since January 2024 (Trading Economics). Average maize retail prices in February 2026 were 15% higher year on year, driven by fuel and transport costs, and up to 20% higher in drought-hit southeastern and coastal areas (FAO GIEWS).

Pastoralists are squeezed from both sides. In Wajir and Mandera, a medium-sized goat bought 55 kg of maize in February 2025 but only 30–35 kg a year later, as goat prices fell and maize prices rose (FAO GIEWS). That is roughly 40% less food for the same animal.

The next shock is already forecast. The Kenya Meteorological Service Authority expects above-average October–December 2026 rains over 80% of the country as a strong El Niño builds (Capital FM, Aug 2026). The IPC already expects El Niño flooding to disrupt access to food in refugee-hosting areas, where needs could rise to 430,000 people by January 2027.

The gap is foresight, not data. WFP's Alert for Price Spikes (ALPS) flags a market only after prices have risen, and WFP warns that its trend line should not be read as a price forecast (WFP ALPS guidance). The Ministry of Agriculture's KAMIS publishes prices from five markets in each county, but it reports today's prices rather than next month's (KAMIS).

The problem is regional. Across six IGAD countries, 40.3 million people face acute food insecurity in 2026, up from 13.9 million in 2016; the regional launch of the Global Report on Food Crises called for stronger early warning and better food security data (The Standard).

Problem statement. County governments, the National Drought Management Authority (NDMA), traders, farmers and households lack timely, market-level warnings of staple food price shocks. Responses such as adjusting cash transfers, releasing strategic reserves, or deciding when to sell or stock therefore come after prices have already spiked.

Stakeholders and users

SokoRadar serves two kinds of users: decision-makers who plan responses, and market actors who decide when to buy, sell or store.

Stakeholder	Decision they make	What SokoRadar gives them	Channel
NDMA and county drought coordinators	When to escalate drought phases and release contingency funds	Market alerts per county, 1–3 months ahead	Dashboard, monthly bulletin
County departments of agriculture and livestock	Market support, livestock offtake, input programmes	Price forecasts and drivers by market	Dashboard
WFP, Kenya Red Cross and NGOs	Cash transfer values, timing of local purchases	Forecasts with uncertainty ranges, open API	Dashboard, API
Ministry of Agriculture and NCPB	Timing of reserve releases and import decisions	National and cross-border price outlook	Dashboard
Smallholder farmers and traders	Sell now, store, or buy stock	Next-month price and alert for their market	USSD, SMS
Pastoralists	When to sell livestock	Goat-to-maize terms of trade outlook	USSD, SMS in Kiswahili

Two personas keep the build focused:

- **Amina, county food security officer in Garissa.** She needs to know which markets will cross into Alert next month, so she can request cash transfer top-ups before prices peak.
- **Wekesa, maize farmer-trader in Bungoma.** After harvest he holds stock and owns a basic phone. He wants one USSD screen that says whether prices are likely to rise or fall over the next two months.

Objectives and research questions

Main objective: build and deploy SokoRadar, a system that forecasts staple food prices one to three months ahead for Kenyan markets and issues explained early warnings through a dashboard, an API and USSD.

Specific objectives

1. Build an automated pipeline that merges monthly market prices with rainfall, vegetation, exchange-rate and global commodity data from 2006 to the present.
2. Forecast monthly retail prices of maize and beans (and goats in pastoral markets) at 1-, 2- and 3-month horizons, beating naive and seasonal-naive baselines.
3. Predict each market's ALPS phase (Normal, Stress, Alert, Crisis) up to three months ahead, prioritising recall on Alert and Crisis.
4. Explain each forecast in plain language by naming its top drivers, such as a rainfall deficit, fuel costs or seasonality.
5. Deploy a web dashboard, a REST API and a USSD/SMS channel, refreshed and retrained automatically every month.
6. Validate usefulness by backtesting the 2011, 2017 and 2022 drought years and testing the USSD flow with at least 10 farmers or traders.

Research questions

1. How far ahead can staple price spikes in Kenyan markets be predicted, and how accurately?
2. Do satellite rainfall and vegetation anomalies improve forecasts beyond price history alone?
3. Which drivers explain price shocks in pastoral (ASAL) markets compared with high-potential farming areas?
4. Does a forecast-based alert warn earlier than WFP's current-month ALPS during past crises?

Data sources

Every core dataset is free, downloadable without approval and updated at least monthly, so the deployed system can keep refreshing itself after the capstone.

Dataset	Provider	What it adds	Coverage	Role in SokoRadar
Kenya – Food Prices	WFP via HDX; sources include NDMA, KNBS, Ministry of Agriculture and EPRA	Monthly retail and wholesale prices by market and commodity	2006 to 2026, monthly updates; CC BY-IGO licence	Target variable and price features

Dataset	Provider	What it adds	Coverage	Role in SokoRadar
Real Time Food Prices – Kenya	World Bank	Gap-filled monthly maize, beans and goat prices	233 markets, Jan 2007 to Aug 2026; only 3–5% of market-months observed, the rest model estimates	Extra features and a benchmark to beat, never ground truth
Rainfall indicators, subnational	WFP and Climate Hazards Center (CHIRPS) via HDX	10-day rainfall, 1- and 3-month totals, anomalies against the 1989–2018 average, short-term rainfall forecasts	Sub-national units, every 10 days	Climate drivers and what-if scenarios
NDVI, subnational	WFP and NASA MODIS via HDX	10-day vegetation index and anomalies	Sub-national units, every 10 days	Crop and pasture condition
Indicative exchange rates	Central Bank of Kenya	Daily KES against USD, UGX and TZS	2003 to present, CSV download	Import and fuel cost pass-through, cross-border prices
Pink Sheet monthly prices	World Bank	Global maize, wheat, crude oil and fertilizer prices	Monthly, updated early each month	Global shock drivers
KAMIS	Ministry of Agriculture	Wholesale, retail and farm-gate prices for 150+ products	Five markets in each of the 47 counties	Checks on recent months and a future live feed, subject to site terms
IPC Kenya analyses	IPC Technical Working Group	County food insecurity phases	Periodic analysis rounds	Validation: do price alerts lead IPC deterioration?

The World Bank series is mostly modelled, so SokoRadar trains and scores only on observed WFP prices; beating the World Bank's own estimates on those prices is a stretch goal.

Methodology

SokoRadar follows CRISP-DM, and every model is judged on one question: does it warn earlier and more accurately than simple baselines?

Data preparation

- Standardise market names, coordinates, commodities and units to KES per kg; keep retail and wholesale series separate.
- Link each market to its county and sub-county so it joins the rainfall and vegetation indicators by admin code.
- Convert 10-day climate data to monthly values and build lags of 1–6 months.
- Keep series with at least 36 monthly prices, WFP's minimum for a reliable seasonal trend; fill short gaps in features only, never in targets.

Feature engineering

- **Price memory:** lags of 1–12 months, rolling means and volatility, year-on-year change.
- **Market network:** prices in the nearest markets and in Nairobi, distance to Nairobi and to the nearest border crossing.
- **Climate:** rainfall and vegetation anomalies for the market's area over the past 1–6 months, plus short-term rainfall forecasts.
- **Macro:** KES/USD rate and global maize, wheat, crude oil and fertilizer prices.
- **Calendar:** month, harvest and lean-season flags for the long and short rains.
- **Pastoral markets:** goat-to-maize terms of trade, the kilograms of maize one goat buys.

Targets

- **Regression:** log retail price at 1, 2 and 3 months ahead for each market and commodity.
- **Classification:** ALPS phase 1–3 months ahead, computed with WFP's method:

ALPS = (observed price − expected price) ÷ residual standard deviation

The expected price comes from regressing price on a time trend and 12 month dummies; the denominator is the standard deviation of the residuals. A market is Normal below 0.25, in Stress from 0.25, Alert from 1 and Crisis from 2 (WFP ALPS guidance). SokoRadar fits the trend on an expanding window, avoiding the look-ahead bias WFP's guidance flags for historical ALPS.

Models

Stage	Models	Purpose
Baselines	Naive (last month), seasonal naive (same month last year)	The bar every model must clear
Statistical	SARIMAX with climate and exchange-rate regressors; Prophet	Interpretable per-market forecasts
Machine learning	Global LightGBM and XGBoost across all markets, one model per horizon, plus quantile versions	Learns shared patterns and handles short series
Deep learning (stretch)	N-BEATS or Temporal Fusion Transformer via Darts	Tests whether deep learning adds value
Alerts	LightGBM classifier with class weights, compared with alerts derived from forecast intervals	Picks the most reliable early-warning route

Uncertainty and explanation

- Conformal prediction (MAPIE) wraps each forecast in an 80% interval with tested coverage; the alert probability is the chance the forecast ALPS reaches 1 or more.
- SHAP values name the top three drivers of each forecast, rewritten as one plain sentence such as "below-normal rains over the last three months".

Validation

- **Rolling-origin backtests:** train on data up to year Y and forecast year Y+1, for every year from 2015 to 2025; 2026 stays untouched as the final test.
- **Crisis replays:** rerun the system month by month through the 2011, 2017 and 2022 drought years and count the months of warning it gave.
- **Benchmarks:** compare against WFP's current-month ALPS and the World Bank estimates on the same observed prices.

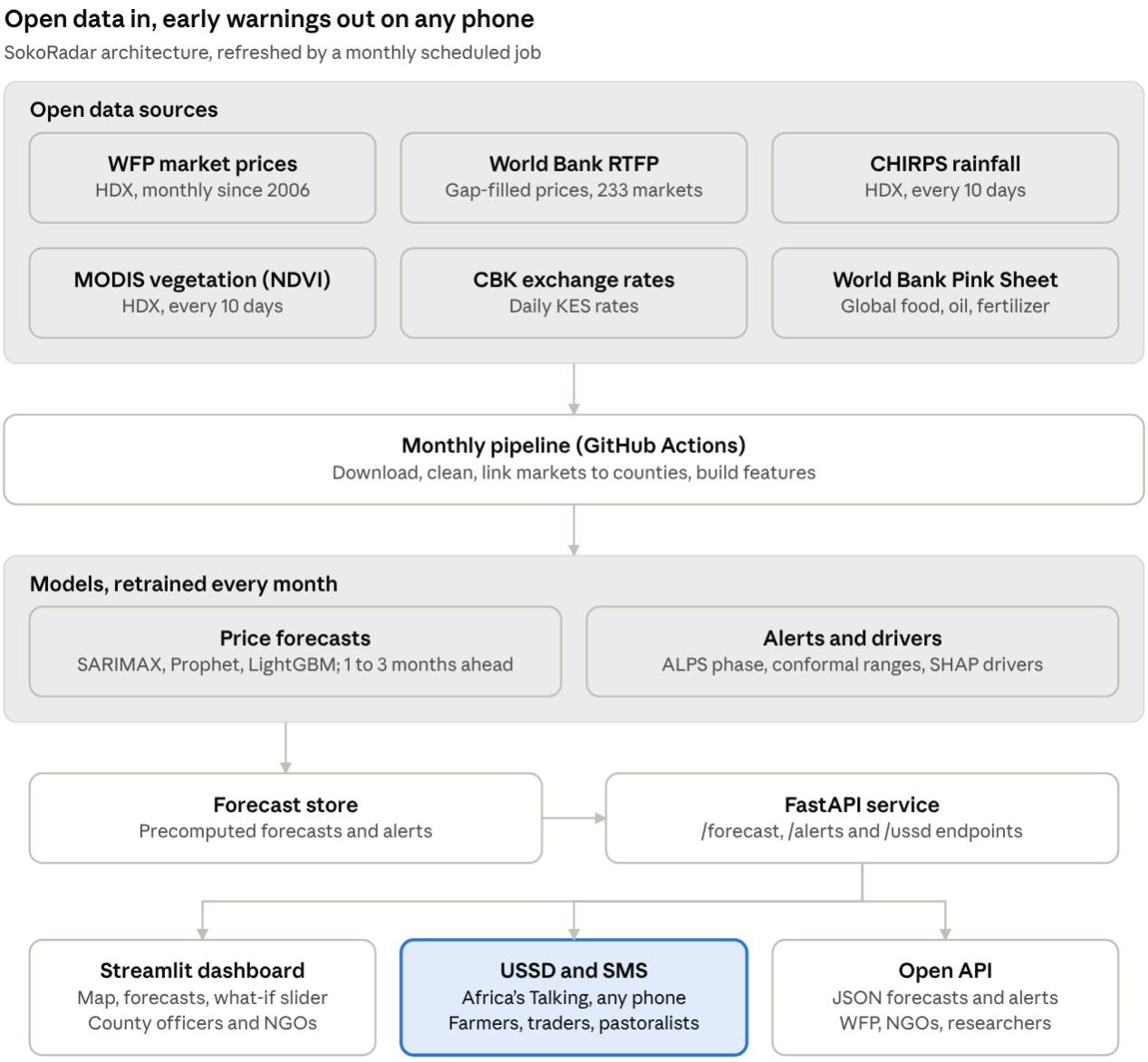
Success metrics

SokoRadar succeeds if it beats simple baselines, warns at least a month earlier than today's alerts, and works on a basic phone.

Area	Metric	Target	Why it matters
Forecasting	MASE against the seasonal-naive forecast	Below 1.0 at all three horizons	Proves the model beats "same month last year"
Forecasting	MAE against the naive (last-month) forecast	At least 10% lower at 2–3 months ahead	Persistence is hard to beat; this shows real foresight
Uncertainty	Share of actual prices inside the 80% interval	75–85%	Ranges users can trust
Alerts	Recall on Alert and Crisis phases	0.70 or more at 1–2 months ahead, with precision of 0.50 or more	A missed crisis costs more than a false alarm
Alerts	Warning lead time in crisis replays	At least one month earlier than current-month ALPS	The core promise of the project
Product	Dashboard load time	Under 3 seconds	Officers use it in meetings
Product	USSD reply time and depth	Under 2 seconds, answer within 3 screens	USSD sessions time out quickly
Product	Unaided USSD task completion	8 of 10 test users	Proves it works for basic-phone users
Operations	Data refresh	Runs end to end from one trigger, no manual steps	The system stays alive after the capstone

Deployment plan and architecture

SokoRadar goes live as three channels — a dashboard, a USSD and SMS service, and an open API — all fed by one monthly pipeline on free hosting.



SokoRadar architecture · 6 open data sources, 3 delivery channels

A scheduled job pulls new data each month, retrains the models and writes forecasts and alerts to a store, so every channel answers from precomputed results.

Component	Tools	Hosting
Data pipeline and retraining	Python, pandas, scheduled GitHub Actions workflow	GitHub
Experiment tracking	MLflow	Local runs, results logged in the repo
Forecast store	Parquet files or a small Postgres database	Repo or a free database tier

Component	Tools	Hosting
API	FastAPI in a Docker container	Render or Hugging Face Spaces
Dashboard	Streamlit with Plotly charts and a county map	Streamlit Community Cloud
USSD and SMS	Africa's Talking sandbox, tested in its browser phone simulator	Free sandbox account

USSD flow (illustrative)

```
CON SokoRadar
1. Bei ya sasa (Price now)
2. Bei ijayo (Next 3 months)
3. Tahadhari (Alerts in my county)
4. Jiandikishe (SMS alerts)

... user picks county, market and commodity ...

END Maize, [market]: KES [price]/kg now.
December: KES [low]-[high], [phase].
Main reason: [top driver].
```

The dashboard adds a what-if slider for rainfall scenarios, a replay mode for 2011, 2017 and 2022, and a data-freshness badge on every market. Each month the pipeline also scores last month's forecasts against actual prices, so accuracy is tracked in the open.

What makes SokoRadar stand out

SokoRadar combines a real national crisis, five open data sources, honest uncertainty and a channel that reaches basic phones — and it can be demonstrated live.

- **Forward-looking, on a scale users already trust.** It forecasts the ALPS phase WFP already uses, so humanitarian users can act on it without learning a new index.
- **Satellites meet markets.** Prices, satellite rainfall, vegetation, exchange rates and global commodity prices run through one pipeline.
- **No smartphone needed.** A USSD menu lets any phone get a forecast; SMS pushes alerts to subscribers.
- **A pastoralist lens.** Forecasting goat-to-maize terms of trade targets the exact squeeze reported in Wajir and Mandera.
- **No black box.** Every forecast carries an interval and its top drivers in plain words.

- **Proven on history.** Replays of the 2011, 2017 and 2022 drought years show how early it would have warned.
- **Timely.** It launches into an El Niño short-rains season, with a what-if slider for rainfall scenarios.
- **Built to last and travel.** A scheduled pipeline keeps it current, and WFP's price database covers 98 countries, so neighbours such as Uganda, Tanzania and Somalia need mostly configuration changes.

Demo-day plan (about 4 minutes):

1. The Kenya map lights up with next month's alerts by market.
2. Replay 2022 month by month and watch when each market is first flagged.
3. A judge dials the USSD code in the simulator and gets a forecast for their chosen market.
4. Move the rainfall slider and watch the alerts change.

Team roles

Five roles cover the build; a four-person team merges Product and Impact into Deployment.

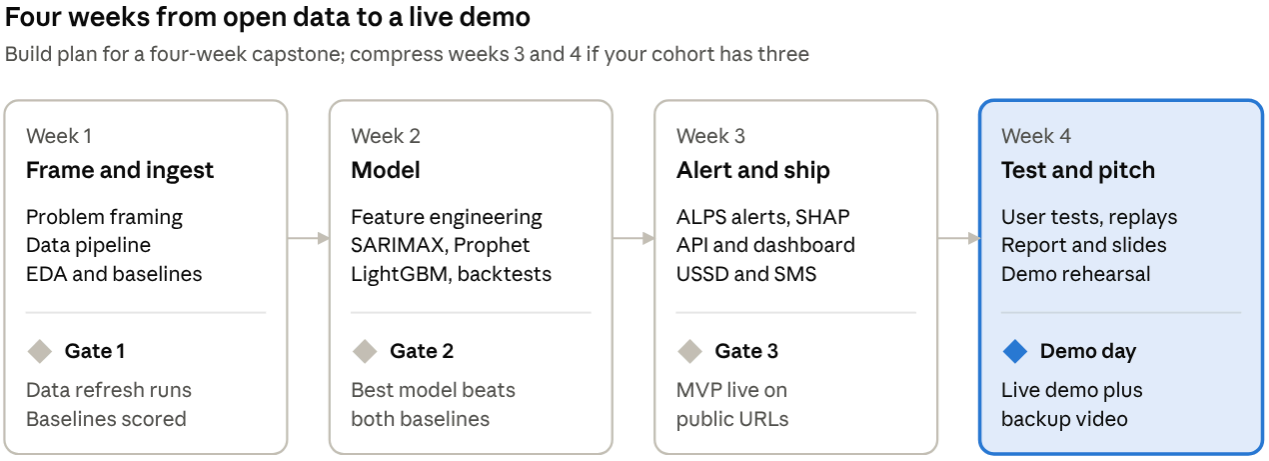
Role	Owns	Key deliverables
Data engineer	Ingestion from HDX, World Bank and CBK; cleaning; scheduled refresh	Clean feature table, data dictionary, data quality report
Forecasting lead	Baselines, SARIMAX and Prophet, global LightGBM, rolling backtests	Model comparison notebook and selected model
Alerts and explainability lead	ALPS targets, alert classifier, conformal intervals, SHAP sentences, crisis replays	Alert model and replay report
Deployment engineer	FastAPI service, Streamlit dashboard, Docker, hosting, monitoring	Live URLs and API documentation
Product and impact lead	USSD and SMS flow, Kiswahili text, user testing, report, pitch and demo script	USSD demo, user-test results, final presentation

Everyone reviews code, writes their README section and presents one part of the demo.

Open: assign a member to each role and add the group name.

Timeline and deliverables

The build runs in four one-week sprints, each ending at a gate the team must pass before moving on.



Build plan · 4 one-week sprints, 4 gates

If a gate slips, the stretch items go first: deep learning, the KAMIS feed and beating the World Bank estimates.

Deliverables

- GitHub repository with README, notebooks, source code, tests and a data dictionary
- Live Streamlit dashboard and FastAPI service on public URLs
- USSD and SMS demo in the Africa's Talking simulator
- Final report and slide deck
- Three-minute backup demo video

Open: confirm the cohort's submission and demo dates so each week can be dated.

Risks, ethics and mitigation

The biggest risks are patchy data in remote markets and forecasts being misread as certainty; both are handled in the design.

Risk	Mitigation
Gaps in remote-market price series	Model only series with 36+ months of data, use World Bank estimates as features, and show data freshness on every forecast
Policy shocks the data cannot see, such as import duty waivers or subsidies	Add policy event flags, widen intervals after shocks, and keep a human review step for published alerts
False alarms or missed crises	Tune for recall on Alert and Crisis, show probabilities and ranges, and never present a forecast as certain
Traders using forecasts to hoard	Publish county alerts openly so all users get the same information at the same time, framed as early warning rather than trading advice
Free hosting sleeps and USSD sessions time out	Precompute forecasts, keep the API warm before demos, and record a backup demo video
KAMIS access or terms change	Treat KAMIS as optional; the core system runs on HDX, World Bank and CBK data
Licensing and attribution	Credit WFP (CC BY-IGO), the World Bank and other sources on the dashboard and in the README
Privacy of SMS subscribers	Store only hashed phone numbers and a county, collect consent, follow Kenya's Data Protection Act 2019, and use the sandbox only during the capstone

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